A closed-form expression for $\mathrm{Tr}\big(M\,(A\otimes A)\big)$ when A is given as a Matrix Product Operator

1 Goal and statement

Fix integers $m \geq 1$, $N \geq 3$, $d \geq 2$, set $D = d^N$, and write $\boldsymbol{P} = \{A, U_1, U_2, 4K\}$ for the four "replica patterns" of $\{a, b, c, d\}$ defined in equation (**??**) below. Let

$$M \in \mathbb{R}^{D^2 \times D^2}$$

be the second-moment matrix of the TT sketch, with rows indexed by $(i^a, i^c) \in [d]^N \times [d]^N$ and columns by (i^b, i^d),

$$M\big[(i^a, i^c),\,(i^b, i^d)\big] \;=\; \mathbb{E}\Big[\,(S^\top S)_{i^a, i^b}\,(S^\top S)_{i^c, i^d}\,\Big]\,. \tag{1}$$

Thanks to the chain construction of the thesis, M admits a description as a finite sum over pattern sequences $\boldsymbol{P}^N$, with per-site pattern transfer matrices W^P and boundary vectors $u_{\mathrm{init}}^{(P_1,P_2)}$, $u_{\mathrm{end}}^{P_N}$, all of which are explicit polynomials in $(m, m_1, m_2, \sigma_2, \sigma_4)$ where $m = m_1 m_2$ is the gadget factorization and (σ_2, σ_4) are the second and fourth moments per entry. We recall this construction in Section **??**.

Suppose now that we are given a symmetric matrix $A \in \mathbb{R}^{D \times D}$ in *Matrix Product Operator (MPO) form*, that is, parameterized by N tensors

$$A^{(l)} \in \mathbb{R}^{r_L^{(l)} \times d \times d \times r_R^{(l)}}, \qquad l = 1, \ldots, N,$$

with $r_L^{(1)} = r_R^{(N)} = 1$ (rank-1 boundaries) and matching internal bonds $r_R^{(l)} = r_L^{(l+1)} =: r_l$. The MPO defines the entries of A via

$$A_{ij} \;=\; A^{(1)}_{i_1 j_1}\, A^{(2)}_{i_2 j_2} \cdots A^{(N)}_{i_N j_N}, \qquad i, j \in [d]^N, \tag{2}$$

where $i = (i_1, \ldots, i_N)$, $j = (j_1, \ldots, j_N)$, and we write $A^{(l)}_{ab} \in \mathbb{R}^{r_{l-1} \times r_l}$ for the matrix obtained by fixing the two physical indices to (a, b). The right-hand side of (**??**) is therefore a product of N matrices of sizes $1 \times r_1, r_1 \times r_2, \ldots, r_{N-1} \times 1$, which evaluates to a scalar.

The goal of this note is to prove and verify the following closed-form expression for the bilinear form $\mathrm{Tr}\big(M(A \otimes A)\big)$.

Theorem 1 (Closed form)**.** *Let M be the matrix defined in* (**??**) *and let A be the matrix defined in* (**??**)*. For every site $l \in \{1, \ldots, N\}$ and every pattern $P \in \boldsymbol{P}$, define the per-site $r_{l-1}^2 \times r_l^2$ matrix*

$$\beta_P^{(l)}\big[(\alpha, \gamma), (\alpha', \gamma')\big] \;=\; \sum_{(a,b,c,d) \in P} A^{(l)}_{ab}[\alpha, \alpha']\, A^{(l)}_{cd}[\gamma, \gamma'], \tag{3}$$

*where the sum is over 4-tuples $(a, b, c, d) \in [d]^4$ that realize the pattern P (Definition **??** below). Then*

$$\boxed{\mathrm{Tr}\big(M\,(A \otimes A)\big) \;=\; \widetilde{u}_{\mathrm{end}}^\top\, \widetilde{W}^{(N-1)}\, \widetilde{W}^{(N-2)} \cdots \widetilde{W}^{(3)}\, \widetilde{u}_{\mathrm{init}}} \tag{4}$$

where the aggregated quantities are

$$\widetilde{W}^{(l)} = \sum_{P\in\boldsymbol{P}} W^P \otimes \big(\beta_P^{(l)}\big)^\top \qquad (l = 3,\dots,N-1)\,, \tag{5}$$

$$\widetilde{u}_{\text{init}} = \sum_{P_1,P_2\in\boldsymbol{P}} u_{\text{init}}^{(P_1,P_2)} \otimes \operatorname{vec}\big(\beta_{P_1}^{(1)}\,\beta_{P_2}^{(2)}\big), \tag{6}$$

$$\widetilde{u}_{\text{end}} = \sum_{P\in\boldsymbol{P}} u_{\text{end}}^{P} \otimes \operatorname{vec}\big(\beta_P^{(N)}\big). \tag{7}$$

The right-hand side of (**??**) is a length-$(N-2)$ matrix-vector product on a space of dimension $4r^2$ (with $r := \max_l r_l$), and is therefore computable in $O(Nr^6)$ floating-point operations. Both $\widetilde{u}_{\text{init}}$ and $\widetilde{u}_{\text{end}}$ are vectors of length $4r^2$.

The remainder of the note proves (**??**). We first introduce notation (Section **??**), recall the chain structure of M (Section **??**), establish the MPO-times-MPO factorization of the joint contraction (Section **??**), and finally combine the two chains into the aggregated form (Section **??**). A numerical verification across six configurations agrees with the brute-force evaluation $\operatorname{vec}(A)^\top M \operatorname{vec}(A)$ to machine precision (Table **??**).

2 Notation and conventions

2.1 Indices and tensor product

We use multi-indices $i = (i_1,\dots,i_N) \in [d]^N$ for the rows and columns of operators on $\mathbb{R}^D$. The Kronecker product of matrices $X \in \mathbb{R}^{m\times n}$ and $Y \in \mathbb{R}^{p\times q}$ is the $mp \times nq$ matrix $X\otimes Y$ with entries

$$(X\otimes Y)\big[(i,k),\,(j,l)\big] \;=\; X_{ij}\,Y_{kl},$$

where the row index (i,k) is read as $i\cdot p+k$ and the column index (j,l) as $j\cdot q+l$. The vectorization $\operatorname{vec}:\mathbb{R}^{p\times q}\to\mathbb{R}^{pq}$ is defined by $\operatorname{vec}(X)_{ip+k} = X_{ik}$ (row-major). Two identities used repeatedly are

$$(X\otimes Y)(Z\otimes W) \;=\; (XZ)\otimes(YW), \qquad \operatorname{Tr}(X\otimes Y) \;=\; \operatorname{Tr}(X)\,\operatorname{Tr}(Y). \tag{8}$$

2.2 The four replica patterns

A *pattern* $P \in \boldsymbol{P}$ is an equivalence relation on the set $\{a,b,c,d\}$ of four "replica labels" such that every block has even size. There are exactly four such relations: the trivial one and three pairings, namely

$$\begin{aligned} 4K &: \{a=b=c=d\},\\ A &: \{a=b\}\sqcup\{c=d\},\\ U_1 &: \{a=c\}\sqcup\{b=d\},\\ U_2 &: \{a=d\}\sqcup\{b=c\}. \end{aligned} \tag{9}$$

Given a 4-tuple $(a,b,c,d)\in[d]^4$, we write $(a,b,c,d)\in P$ to mean that the equality pattern of (a,b,c,d) equals P. Explicitly,

$$\begin{aligned} (a,b,c,d)\in 4K &\iff a=b=c=d,\\ (a,b,c,d)\in A &\iff a=b,\ c=d,\ a\neq c,\\ (a,b,c,d)\in U_1 &\iff a=c,\ b=d,\ a\neq b,\\ (a,b,c,d)\in U_2 &\iff a=d,\ b=c,\ a\neq b. \end{aligned}$$

Tuples outside these four patterns have an odd-sized block and contribute zero to all moments below; we ignore them.

3 Chain structure of M

We recall the form of M produced by the thesis construction. (No new material; this section fixes notation that is used throughout.)

Proposition 2 (Thesis chain decomposition)**.** *There exist scalar quantities $V(P_1, \dots, P_N) \in \mathbb{R}$ indexed by pattern sequences $\boldsymbol{P}^N$ such that, for every choice of multi-indices $i^a, i^b, i^c, i^d \in [d]^N$,*

$$M\big[(i^a, i^c), (i^b, i^d)\big] \;=\; \sum_{(P_1,\dots,P_N)\in \boldsymbol{P}^N} \Big(\textstyle\prod_{l=1}^{N} \mathbf{1}\big[(i_l^a, i_l^b, i_l^c, i_l^d) \in P_l\big]\Big)\, V(P_1, \dots, P_N). \tag{10}$$

Moreover, V has the chain (transfer-matrix) factorization

$$V(P_1, \dots, P_N) \;=\; \big(u_{\mathrm{end}}^{P_N}\big)^{\top} W^{P_{N-1}} W^{P_{N-2}} \cdots W^{P_3}\, u_{\mathrm{init}}^{(P_1,P_2)}, \tag{11}$$

where $W^P \in \mathbb{R}^{4\times 4}$ for $P \in \boldsymbol{P}$ and $u_{\mathrm{init}}^{(P_1,P_2)}, u_{\mathrm{end}}^{P_N} \in \mathbb{R}^4$ are the explicit polynomials in $(m, m_1, m_2, \sigma_2, \sigma_4)$ given by formulas (19), (20), (21) of the thesis.

The interpretation of (**??**) is straightforward: M is built by summing, over all 4^N assignments of a per-site pattern, the product of an indicator (only those (i^a, i^b, i^c, i^d) that match the pattern contribute) and a chain value V.

4 Goal as an explicit four-fold sum

The bilinear form $\mathrm{Tr}(M \cdot Y)$ for any $Y \in \mathbb{R}^{D^2\times D^2}$ is

$$\mathrm{Tr}(MY) \;=\; \sum_{I,J} M_{IJ}\, Y_{JI},$$

where $I, J \in [D^2]$. With $Y = A \otimes A$, decoded into the user's (i^a, i^b, i^c, i^d) convention,

$$(A\otimes A)\big[(i^b, i^d), (i^a, i^c)\big] \;=\; A_{i^b i^a}\, A_{i^d i^c} \;=\; A_{i^a i^b}\, A_{i^c i^d} \quad (\text{since } A = A^{\top}).$$

Hence

$$\mathrm{Tr}\big(M(A\otimes A)\big) \;=\; \sum_{i^a,i^b,i^c,i^d\in[d]^N} A_{i^a i^b}\, A_{i^c i^d}\, M\big[(i^a, i^c), (i^b, i^d)\big]. \tag{12}$$

This identity is independent of any sketch structure and holds for any symmetric A and any M in the user's convention.

5 The MPO factorization of $A_{i^a i^b}\, A_{i^c i^d}$

Substituting (**??**) into the product yields, with double indices on the bond legs $(\alpha_l)_{l=0}^N$ and $(\gamma_l)_{l=0}^N$ (both with closed boundaries $\alpha_0 = \alpha_N = \gamma_0 = \gamma_N = 1$),

$$A_{i^a i^b}\, A_{i^c i^d} \;=\; \Big(\prod_{l=1}^{N} A^{(l)}_{i_l^a i_l^b}[\alpha_{l-1}, \alpha_l]\Big)\, \Big(\prod_{l=1}^{N} A^{(l)}_{i_l^c i_l^d}[\gamma_{l-1}, \gamma_l]\Big).$$

Combining the two products site-by-site as a tensor product on the bond legs,

$$A_{i^a i^b}\, A_{i^c i^d} \;=\; \prod_{l=1}^{N} \big(A^{(l)}_{i^a_l i^b_l} \otimes A^{(l)}_{i^c_l i^d_l}\big)\big[(\alpha_{l-1},\gamma_{l-1}),\,(\alpha_l,\gamma_l)\big], \tag{13}$$

where the product is now a matrix product on the joint bond space $\mathbb{R}^{r_{l-1}^2} \to \mathbb{R}^{r_l^2}$, evaluated at the closed boundary indices.

6 Combining the two chains

We now substitute (**??**) and (**??**) into (**??**) and exchange sums.

6.1 Per-site collapse via $\beta_P^{(l)}$

Plugging (**??**) into (**??**):

$$\mathrm{Tr}(M(A\otimes A)) \;=\; \sum_{(P_1,\ldots,P_N)} V(P_1,\ldots,P_N) \sum_{\substack{i^a,i^b,i^c,i^d:\\ (i^a_l,i^b_l,i^c_l,i^d_l)\in P_l\,\forall l}} A_{i^a i^b}\, A_{i^c i^d}.$$

The inner sum factorizes site by site because the indicator constraints and the MPO product (**??**) are both products over l:

$$\begin{aligned}
&\sum_{\substack{i^a,i^b,i^c,i^d:\\ (i^a_l,i^b_l,i^c_l,i^d_l)\in P_l\,\forall l}} A_{i^a i^b}\, A_{i^c i^d} \\
&\quad= \sum_{\alpha,\gamma} \prod_{l=1}^{N} \sum_{(a,b,c,d)\in P_l} \big(A^{(l)}_{ab} \otimes A^{(l)}_{cd}\big)\big[(\alpha_{l-1},\gamma_{l-1}),(\alpha_l,\gamma_l)\big] \\
&\quad= \sum_{\alpha,\gamma} \prod_{l=1}^{N} \beta^{(l)}_{P_l}\big[(\alpha_{l-1},\gamma_{l-1}),(\alpha_l,\gamma_l)\big],
\end{aligned}$$

where in the last step we used the definition (**??**) of $\beta_P^{(l)}$. The remaining sum over closed bond indices (α,γ) is, by inspection, the $(1,1)$-entry of the matrix product of β's:

$$\sum_{\substack{i^a,i^b,i^c,i^d:\\ (i^a_l,i^b_l,i^c_l,i^d_l)\in P_l\,\forall l}} A_{i^a i^b}\, A_{i^c i^d} \;=\; \big(\beta^{(1)}_{P_1}\,\beta^{(2)}_{P_2}\cdots\beta^{(N)}_{P_N}\big)[1,1]. \tag{14}$$

(Recall that $r_0 = r_N = 1$, so this product has shape 1×1.)

6.2 Joining the two chains

Substituting (**??**) and (**??**):

$$\mathrm{Tr}(M(A\otimes A)) \;=\; \sum_{(P_1,\ldots,P_N)} \Big[\,\big(u^{P_N}_{\mathrm{end}}\big)^{\top} W^{P_{N-1}}\cdots W^{P_3}\, u^{(P_1,P_2)}_{\mathrm{init}}\Big]\Big[\,\beta^{(1)}_{P_1}\,\beta^{(2)}_{P_2}\cdots\beta^{(N)}_{P_N}\Big][1,1]. \tag{15}$$

The two bracketed factors are independent products: the first runs over $P_3, \dots, P_{N-1}$ in the 4-dimensional "replica-pattern" space, the second runs over $P_1, \dots, P_N$ in the r^2-dimensional "MPO-bond" space. Because the patterns at different sites are summed independently, we can interleave the sums: at each interior site $l \in \{3, \dots, N-1\}$ the contribution is

$$\sum_{P_l \in \boldsymbol{P}} W^{P_l} \;\otimes\; \beta_{P_l}^{(l)\top}.$$

The transpose appears because the W^{P_l} is acting on the replica space in the "out $\leftarrow$ in" direction (left multiplication), whereas $\beta_{P_l}^{(l)}$, by definition (**??**), has its rows indexed by the *left* bond and its columns by the *right* bond, and thus acts on the bond space in the "in $\rightarrow$ out" direction (right multiplication). Transposing $\beta_{P_l}^{(l)}$ aligns the two directions so they can be combined into a single "out $\leftarrow$ in" transfer matrix

$$\widetilde{W}^{(l)} \;=\; \sum_{P \in \boldsymbol{P}} W^{P} \otimes \big(\beta_{P}^{(l)}\big)^{\top} \quad \in\; \mathbb{R}^{4r_l^2 \times 4r_{l-1}^2}, \tag{16}$$

which matches (**??**).

The boundary terms are obtained by absorbing sites $l \in \{1, 2\}$ into a single initial vector and site $l = N$ into a single end vector. For the initial vector, sites 1 and 2 contribute

$$u_{\text{init}}^{(P_1,P_2)} \;\otimes\; \mathrm{vec}\big(\beta_{P_1}^{(1)}\, \beta_{P_2}^{(2)}\big),$$

where $\mathrm{vec}(\,\cdot\,)$ flattens the $1 \times r_2^2$ row vector into an element of $\mathbb{R}^{r_2^2}$ (the leftmost MPO bond is closed, $r_0 = 1$, so $\beta_{P_1}^{(1)}\, \beta_{P_2}^{(2)}$ is a row vector). For the end vector, site N contributes

$$u_{\text{end}}^{P_N} \;\otimes\; \mathrm{vec}\big(\beta_{P_N}^{(N)}\big),$$

where $\beta_{P_N}^{(N)}$ is a column vector of length r_{N-1}^2 (rightmost bond closed, $r_N = 1$). Summing over P_1, P_2, P_N gives the expressions (**??**) and (**??**).

The total contraction is therefore the matrix-vector chain (**??**), which proves Theorem **??**.

6.3 The variance

A small corollary, useful in applications: since $\mathbb{E}[S^\top S] = I_D$,

$$\mathbb{E}\big[\langle S^\top S, A\rangle\big] \;=\; \mathrm{tr}(A),$$

and consequently

$$\mathrm{Var}\big(\langle S^\top S, A\rangle\big) \;=\; \mathrm{Tr}\big(M\,(A \otimes A)\big) \;-\; \mathrm{tr}(A)^2, \tag{17}$$

where $\mathrm{Tr}(M(A \otimes A))$ is given by (**??**) and $\mathrm{tr}(A)$ is the standard MPO trace

$$\mathrm{tr}(A) \;=\; \prod_{l=1}^{N} \Big(\sum_{a \in [d]} A_{aa}^{(l)} \Big).$$

7 Closed-form values of $\beta_P^{(l)}$

It is convenient to write each $\beta_P^{(l)}$ in terms of simpler bond matrices. Let

$$S_d^{(l)} \;=\; \sum_{a=1}^{d} A_{aa}^{(l)}, \qquad \Delta^{(l)} \;=\; \sum_{a=1}^{d} A_{aa}^{(l)} \otimes A_{aa}^{(l)}.$$

Then (**??**) unfolds as follows.

$$\begin{aligned}
\beta_{4K}^{(l)} &= \Delta^{(l)}, \\
\beta_{A}^{(l)} &= \big(S_d^{(l)}\big) \otimes \big(S_d^{(l)}\big) \;-\; \Delta^{(l)}, \\
\beta_{U_1}^{(l)} &= \sum_{a,b=1}^{d} A_{ab}^{(l)} \otimes A_{ab}^{(l)} \;-\; \Delta^{(l)}, \\
\beta_{U_2}^{(l)} &= \sum_{a,b=1}^{d} A_{ab}^{(l)} \otimes A_{ba}^{(l)} \;-\; \Delta^{(l)}.
\end{aligned}$$

Sanity checks.

- For $A^{(l)} = I_d$ (the identity, an MPO of rank 1), one has $A_{ab}^{(l)} = \delta_{ab}$, so $\Delta^{(l)} = d$ and $\beta_{4K}^{(l)} = d$, $\beta_A^{(l)} = d^2 - d = d(d-1)$, $\beta_{U_1}^{(l)} = \beta_{U_2}^{(l)} = 0$. These are exactly the multiplicities used in the trace lemma of the thesis (the case $A = I$ of (**??**) reduces to $\mathrm{Tr}(M)$).
- For $A^{(l)} = J_d = \mathbf{1}\mathbf{1}^\top$, one has $A_{ab}^{(l)} = 1$ for all a, b, so $\beta_{4K}^{(l)} = d$ and $\beta_A^{(l)} = \beta_{U_1}^{(l)} = \beta_{U_2}^{(l)} = d(d-1)$. This recovers the all-ones A case discussed in the thesis.

8 Numerical verification

We compute (**??**) and compare against the brute-force evaluation $\mathrm{vec}(A)^\top M\ \mathrm{vec}(A)$ for the materialized $D^2 \times D^2$ matrix M. Random MPO cores $A^{(l)}$ are drawn i.i.d. Gaussian with shape $r_{l-1} \times d \times d \times r_l$, varying the bond rank r. Six configurations:

m	N	d	r	dist.	$\mathrm{Tr}(M(A \otimes A))$ (closed)	$\mathrm{vec}(A)^\top M\ \mathrm{vec}(A)$	rel. err.
4	3	2	1	gauss	1.20357×10^{1}	1.20357×10^{1}	8.9×10^{-16}
4	3	2	2	gauss	2.62746×10^{2}	2.62746×10^{2}	6.5×10^{-16}
4	4	2	2	rad	2.10802×10^{2}	2.10802×10^{2}	2.7×10^{-16}
4	4	2	3	gauss	3.93434×10^{4}	3.93434×10^{4}	1.9×10^{-16}
4	5	2	2	rad	1.02678×10^{5}	1.02678×10^{5}	2.8×10^{-16}
6	4	3	2	gauss	2.68329×10^{4}	2.68329×10^{4}	4.1×10^{-16}

Table 1: Closed form vs. brute force for $\mathrm{Tr}(M(A \otimes A))$, agreement to the level of double-precision rounding.

A reference implementation in approximately 100 lines of Python is provided in the file `trace_MB_mpo.py`; running it as a script reproduces the table above.

9 Cost and remarks

Computational cost. With $r := \max_l r_l$, the dominant cost is the propagation step (**??**): each $\widetilde{W}^{(l)}$ has size $4r^2 \times 4r^2$, and its application to a vector of length $4r^2$ costs $O(r^4)$ floating-point operations (after exploiting the Kronecker structure, the cost is $O(r^4)$ and not $O(r^8)$). Over $N-3$ interior sites, total cost is $O(Nr^4)$. Building each $\beta_P^{(l)}$ from the core $A^{(l)}$ costs $O(d^2r^4)$.

Comparison with the row-sum bound. The thesis row-sum lemma gives $\mathrm{Tr}(MB) \le R \cdot \mathrm{Tr}(B)$ where R is the maximum row sum of M. For PSD A this implies $\mathrm{Var}(\langle S^\top S, A\rangle) \le (R-1)\ \mathrm{tr}(A)^2$. Numerically $R-1$ scales as $\alpha(d)^N/m$ with $\alpha(d) \approx d$, so the bound is exponential in N. The closed form (**??**) replaces the bound by an exact value and exhibits no such blow-up; in particular for $A = I$ one verifies $\mathrm{Tr}(M)/d^{2N} \approx (1 + a(d)/m)^N$ with $a(d)$ a small constant, which is the optimal-shape regime.

Higher moments. The same construction extends to $\mathrm{Tr}\big((M(A\otimes A))^k\big)$ for any $k \ge 1$ by treating the k-fold contraction as a chain of k stacked aggregated transfer matrices, with appropriate trace-closure boundary. The result is a matrix-vector chain on $\mathbb{R}^{(4r^2)^k}$; we omit the details here.

A Reminder: explicit forms of W^P, $u_{\mathrm{init}}^{(P_1,P_2)}$, $u_{\mathrm{end}}^{P_N}$

For completeness we record the formulas from the thesis. With $\sigma_2 = 1/m$, and $\sigma_4 = \sigma_2^2$ (Rademacher) or $\sigma_4 = 3\sigma_2^2$ (Gaussian), and (m_1, m_2) the closest integer factor pair of m, define for $P, Q \in \boldsymbol{P}$ the partition meet $P \wedge Q$ (the partition whose blocks are intersections of P- and Q-blocks; if $P \wedge Q$ has any odd-sized block, set $\pi(P \wedge Q) := \varnothing$ and let the moments below vanish). Define the multiplicity

$$N_P(M) \;=\; \begin{cases} M & \text{if } P = 4K, \\ M(M-1) & \text{otherwise,} \end{cases}$$

and the moment functions

$$\mu^X_{P,B} \;=\; \begin{cases} \sigma_4 & \pi(P \wedge B) = 4K, \\ \sigma_2^2 & \pi(P \wedge B) \in \{A, U_1, U_2\}, \\ 0 & \text{else,} \end{cases} \qquad \mu^g_{L,R} \;=\; \sum_{s \in \boldsymbol{P}} N_s(m)\, \mu^X_{L,s}\, \mu^X_{R,s}.$$

The 4×4 pattern transfer matrix is

$$W^P\big[Z_o, Z_i\big] \;=\; N_{Z_o}(m) \sum_{s_1, s_2 \in \boldsymbol{P}} N_{s_1}(m_1)\, N_{s_2}(m_2)\, \mu^X_{P,\, s_1 \wedge s_2}\, \mu^g_{Z_i \wedge s_1,\, s_2 \wedge Z_o}, \tag{18}$$

and the boundary vectors $u_{\mathrm{init}}^{(P_1,P_2)}, u_{\mathrm{end}}^{P_N}$ are given by analogous explicit polynomials (formulas (20), (21) of the thesis). These quantities depend only on m and the noise distribution, not on A, N, or d.